

TMS9918B/TMS9928B/TMS9929B

Video Display Processors

Data Manual

Pin-for-pin replacement for the TMS9918A/TMS9928A/TMS9929A

Same crystal, same 16K x 1 dynamic VRAM board (120 ns, page mode)

Extended modes: 2 bpp tiles with palettes, bitmap, 64-column text

Eight sprites per line, hardware scrolling, scanline interrupt

ADVANCE INFORMATION

Hypothetical 1983 device - design study prepared for the GearSF7000 emulator.

Not a Texas Instruments product. TMS9918A is a trademark reference used for compatibility only.

September 2026

IMPORTANT NOTICES

This manual describes the TMS9918B, TMS9928B and TMS9929B, a hypothetical revision of the Texas Instruments TMS9918A family developed as a design study. Every electrical and timing value marked as derived was computed from the data sheets of the dynamic RAMs fitted to 1983 Sega SC-3000 boards and from the TMS9918A/9928A/9929A Video Display Processors Data Manual (MP010A). Values not restated here are those of MP010A.

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The design rationale, cost analysis and rejected alternatives are in the separate TMS9918B Design Notes. Programming examples are in the TMS9918B Programmer's Guide Supplement.

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1. INTRODUCTION

1.1 Description

The TMS9918B/9928B/9929B video display processors (VDP) are N-channel MOS LSI devices that generate all video, control and synchronization signals of a raster-scanned colour television or monitor and control the storage, retrieval and refresh of display data in a dynamic screen refresh memory. They are pin-for-pin replacements of the TMS9918A/9928A/9929A and use the same 10.738635 MHz crystal and the same eight 16K x 1 dynamic RAMs.

After reset the TMS9918B behaves as a TMS9918A: the same registers, the same four display modes and undocumented modes, the same VRAM access sequence and the same CPU access windows. Software unlocks the extended functions with a register command. The extended functions add a 16-entry palette of TMS colours with four luminance levels, six extended display modes, eight sprites per line with three-colour sprite pairs, hardware scrolling and a scanline interrupt.

The VDP reads VRAM with a memory cycle placed on half-periods of the crystal and fetches adjacent bytes in page mode. The extended functions therefore require dynamic RAMs with 120 ns access time and page-mode capability (Section 3.1 and Appendix A).

The TMS9928B/9929B are functionally identical to the TMS9918B except for the colour encoding, which is replaced by luminance and colour-difference outputs; the TMS9929B uses the 625-line format.

1.2 Features

- Pin-for-pin and software compatible with the TMS9918A/9928A/9929A
- Same crystal (10.738635 MHz) and same 16K x 1 dynamic VRAM (120 ns, page mode)
- 256 x 192 resolution; 512-pixel lines in 64-column text
- 15 TMS colours at 4 luminance levels through a 16-entry write-only palette
- Extended modes: Graphics1X (512 two-bit tiles), Graphics2Fat (16 colours per 2-pixel dot), linear Bitmap, Text 40 with colours per character, Text 64 with 768 characters
- Eight sprites per line, 512 sprite patterns, three-colour sprite pairs, tile priority over sprites
- Horizontal and vertical hardware scroll with left-column blanking and locked areas
- Scanline interrupt with programmable line count
- Faster CPU access: 14 to 18 T-states worst case in extended modes (29 T in Graphics mode)
- Standard 40-pin package

1.3 Typical Applications

- Home computers and video game consoles based on the TMS9918A family
- Colour terminals with 64-column text
- Home educational systems
- European 625-line TV (TMS9929B)

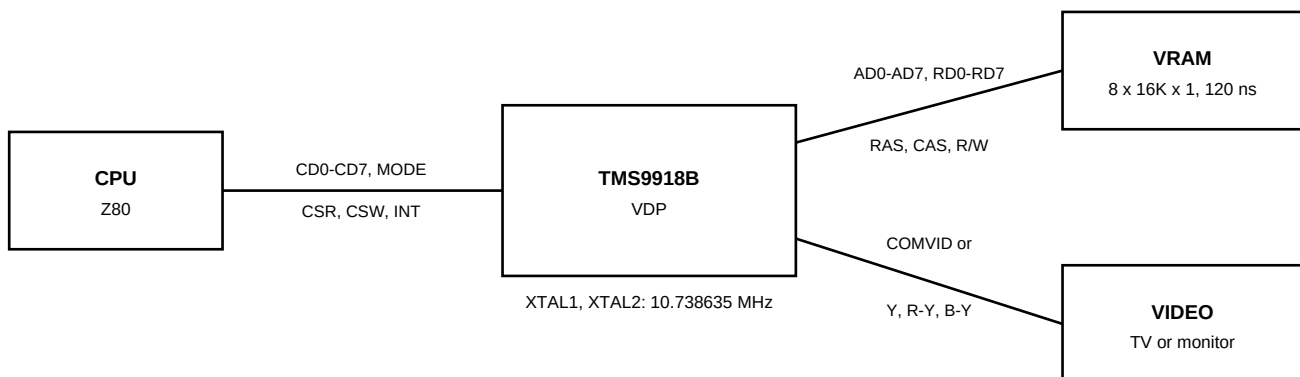


FIGURE 1-1 - SYSTEM BLOCK DIAGRAM

1.4 Acronyms and Glossary

Term	Meaning
Unit	Half a period of the 10.738635 MHz crystal, 46.56 ns. Timing positions in this manual are given in units.
Slot	TMS9918A memory cycle position, 8 units (372 ns). 171 slots per line.
Burst	Page-mode memory cycle reading 2 to 4 adjacent bytes under one RAS pulse.
Extended mode	Display mode selected with MX = 1 after the unlock command (Section 2.5).
Palette entry	6-bit value: TMS colour code and luminance level.
Stratum	One bit plane of a 2-bit-per-pixel pattern.
T-state	Z80 clock period at 3.58 MHz, 6 units.
VRAM	Video refresh memory, 16384 bytes of dynamic RAM.

2. ARCHITECTURE

2.1 CPU Interface

The CPU interface is identical to the TMS9918A: an 8-bit bidirectional bus (CD0-CD7, CD0 is the MSB), MODE, CSR, CSW and INT. Bit 0 is the most significant bit throughout this manual, as in MP010A.

2.1.1 Control port codes

A two-byte control transfer carries an address or a register value in the first byte and a code in the two most significant bits of the second byte.

Second byte D0 D1	TMS9918A	TMS9918B, XE = 0	TMS9918B, XE = 1
0 0	VRAM read address	VRAM read address	VRAM read address
0 1	VRAM write address	VRAM write address	VRAM write address
1 0	register write	register write	register write
1 1	register write	register write	palette write address (first byte D4-D7 = entry)

TABLE 2-1 - CONTROL PORT CODES

2.1.2 Register unlock

After reset the register number decodes three bits, as on the TMS9918A: register 8 aliases R0 and register 15 aliases R7. Two consecutive register writes of **5Ah** to register 63 (second byte **BFh**), without another register write in between, unlock registers 8 to 15. Only RESET locks them again. While locked the command lands in R7; software rewrites R7 afterwards.

NOTE

Register 63 and the value 5Ah do not collide with the F18A/PICO9918 unlock (1Ch to register 57, which lands in R1 on a locked TMS9918B) nor with R15, the status register select of the V9938 and F18A.

2.1.3 CPU access to VRAM

The 14-bit autoincrementing address register, the read-ahead byte and the transfer sequences are those of the TMS9918A. The VDP performs a CPU transfer in the next memory cycle reserved for the CPU; worst-case delays for every mode are listed in Appendix B.

2.1.4 VDP interrupt

INT is active (low) when F = 1 and IE = 1 (frame interrupt, as on the TMS9918A) or when FL = 1 and IE1 = 1 (scanline interrupt, Section 2.7). Reading S0 clears F; reading S1 clears FL.

2.1.5 VDP initialization

RESET clears R0, R1, R8 and R13, locks the extended registers and selects status register S0. The palette and R9 to R15 are undefined after power-up and must be written before extended modes are used.

2.2 Write-Only Registers

Registers R0 to R7 keep their TMS9918A functions (MP010A Section 2.2). Registers R8 to R15 exist after the unlock command. Figure 2-1 shows the extended registers; reserved bits must be written as 0.

R8 MODE	D0	D1	D2	D3	D4	D5	D6	D7
	IE1	0	0	0	0	0	MX	XE
R9 H SCROLL	D0	D1	D2	D3	D4	D5	D6	D7
	COLUMNS					FINE		
R10 V SCROLL	D0	D1	D2	D3	D4	D5	D6	D7
	LINES (0-191)							
R11	D0	D1	D2	D3	D4	D5	D6	D7
	RESERVED							

	D0	D1	D2	D3	D4	D5	D6	D7
R12 LINE COUNT	RELOAD VALUE							
	D0	D1	D2	D3	D4	D5	D6	D7
R13 SCREEN	0	0	0	0	HLOCK	VLOCK	MASK	T64
	D0	D1	D2	D3	D4	D5	D6	D7
R14	RESERVED							
	D0	D1	D2	D3	D4	D5	D6	D7
R15 STATUS SEL	0	0	0	0	STATUS REGISTER NUMBER			

FIGURE 2-1 - EXTENDED REGISTERS (D0 = MSB)

Register / bit	Function
R8 D7 XE	Extended enable: palette port, S1, R9-R15 and IE1 become active. With XE = 0 the VDP is a TMS9918A.
R8 D6 MX	Extended display modes (requires XE). Selects the modes of Table 2-3 together with M1, M2, M3.
R8 D0 IE1	Scanline interrupt enable.
R9	World X = screen X + R9 (modulo 256). D0-D4 whole columns, D5-D7 pixels.
R10	World line = (line + R10) modulo 192.
R12	Scanline interrupt reload value (Section 2.7).
R13 D7 T64	Extended text modes use 64 columns instead of 40.
R13 D6 MASK	The leftmost 8 pixels show the backdrop colour, sprites included.
R13 D5 VLOCK	Columns 24-31 do not scroll vertically.
R13 D4 HLOCK	Lines 0-15 do not scroll horizontally.
R15 D4-D7	Status register returned by a status read: 1 selects S1, any other value selects S0.

TABLE 2-2 - EXTENDED REGISTER FUNCTIONS

NOTE

In extended modes R3 is not used, R2 and R4 locate the extended tables (Section 3.2) and R7 supplies the backdrop colour and the text colours. R5 and R6 keep their TMS9918A meaning.

2.3 Status Registers

	D0	D1	D2	D3	D4	D5	D6	D7
S0	F	5S	C	SPRITE NUMBER				
	D0	D1	D2	D3	D4	D5	D6	D7
S1	0	0	0	1	1	0	0	FL

FIGURE 2-2 - STATUS REGISTERS

S0 is the TMS9918A status register. In extended graphics modes 5S reports the ninth sprite found on a line pair (Section 2.6). S1 returns the identification value 18h plus FL (01h), the scanline interrupt flag, cleared when S1 is read. The identification fails the published detection masks of the F18A (111xxxxx), PICO9918 (E8h), V9938 and V9958, and reads as 18h when masked with 3Eh.

2.4 Palette

	D0	D1	D2	D3	D4	D5	D6	D7
ENTRY	0	0	LUMINANCE		TMS COLOUR CODE			

FIGURE 2-3 - PALETTE ENTRY

The palette holds 16 write-only entries on the chip: four palettes of four entries. The colour code selects one of the 15 TMS9918A colours; the luminance level scales the video output above black level: 00 full, 01 3/4, 10 1/2, 11 1/4. A

palette write sets the entry number with control code 11 (Table 2-1); every following data port write stores one entry and increments the entry number modulo 16. Palette writes do not wait for a memory cycle.

Entry 0 of each palette is stored but never displayed: pixel value 0 shows the backdrop (R7 D4-D7) in playfield modes and is transparent for sprites.

2.5 Video Display Modes

XE MX	M3	M2	M1	Mode	Resolution	Sprites/line
0 x or 1 0	-	-	-	TMS9918A modes, documented and undocumented	-	4 (Text: 0)
1 1	0	0	0	Graphics1X	256 x 192, 32 x 24 tiles, 512 patterns, 2 bpp	8
1 1	1	0	0	Graphics2Fat	128 x 192 (2-pixel dots), 768 patterns	8
1 1	0	0	1	Text40X / Text64 (T64 = 1)	40 x 24 (6 x 8) / 64 x 24 (8 x 8)	cursor
1 1	1	0	1	Text40XQ / Text64Q	as above, up to 768 characters	cursor
1 1	0	1	0	Bitmap	256 x 192, 2 bpp	8
1 1	1	1	0	BitmapQ	256 x 192, 2 bpp, third banking	8
1 1	x	1	1	bars (as TMS9918A)	-	0

TABLE 2-3 - DISPLAY MODES (M1 = R1 D3, M2 = R1 D4, M3 = R0 D6)

NOTE

With XE = 0 or MX = 0 all eight M1-M2-M3 combinations behave as on the TMS9918A, including the undocumented Text 1Q (M1 + M3), Multicolor Q (M2 + M3) and bar modes (M1 + M2, M1 + M2 + M3), with the TMS9918A table addresses and memory access sequences: Text for M1 = 1, Multicolor for M2 = 1 and M1 = 0, Graphics otherwise.

M3 selects the third-banking variant of each extended mode: R4 D6-D7 give the middle and bottom thirds of the screen their own table blocks, as in Graphics II.

2.5.1 Graphics1X Mode

Each of the 768 tile positions has a 2-byte name entry: the pattern name and an attribute. A pattern has 16 bytes, two for each of its eight rows (stratum 0, stratum 1). Pixel value = stratum 0 bit + 2 x stratum 1 bit; value 0 is the backdrop and values 1-3 are entries of the palette selected by the attribute.

ATTRIBUTE	D0	D1	D2	D3	D4	D5	D6	D7
	0	0	0	0	NAME8	PRIOR	PALETTE	

FIGURE 2-4 - GRAPHICS1X ATTRIBUTE

NAME8 is the ninth bit of the pattern name (512 patterns). PRIOR places the non-backdrop pixels of the tile in front of all sprites.

2.5.2 Graphics2Fat Mode

The name entry and the 16-byte pattern layout are those of Graphics1X, with third banking (768 patterns) and the attribute PRIOR bit only. Each pattern row describes four dots, each two pixels wide: dot p takes bits (2p, 2p+1) of stratum 0 as its low bits and the same bits of stratum 1 as its high bits (D0 = MSB). The 4-bit value is a TMS colour code at full luminance; 0 is the backdrop.

2.5.3 Bitmap and BitmapQ Modes

The bitmap stores two bytes (stratum 0, stratum 1) for every 8 pixels of every line: 64 bytes per line, 12288 bytes for the screen. A palette map holds one byte for every 8 x 8 area; its D6-D7 select the palette of the area. In BitmapQ each third of the screen uses a 4096-byte block selected as in Graphics II.

2.5.4 Text40X and Text64 Modes

Text40X keeps the 40 x 24 layout of Text mode with a 2-byte entry per character: the pattern name and a colour byte whose D0-D3 give the foreground and D4-D7 the background; a zero nibble uses the corresponding nibble of R7, so a colour byte of 00h displays exactly as Text mode.

With $T64 = 1$ the VDP displays 64 x 24 characters of 8 x 8 pixels on 512-pixel lines (one name byte per character, all 8 pattern bits used). The two colours of the whole screen come from R7. Monitors with RGB or component input are required for legible 64-column text.

In both text modes sprites 0 and 1 are displayed as a hardware cursor (a pair when sprite 1 has PAIR set); in 64-column text the sprite X coordinate is doubled.

2.6 Sprites

The Sprite Attribute Table and Sprite Pattern Table keep the TMS9918A layout. In extended graphics modes the fourth attribute byte has the format of Figure 2-5 and up to eight sprites are displayed per line.

COLOUR BYTE	D0	D1	D2	D3	D4	D5	D6	D7
	EC	0	BANK	PAIR	PALETTE		ENTRY	

FIGURE 2-5 - SPRITE ATTRIBUTE BYTE 3 IN EXTENDED MODES

Field	Function
EC	Early clock: $X - 32$, as TMS9918A.
BANK	Pattern taken from the 2048-byte table that follows the one located by R6.
PAIR	Odd sprites only: sprite $2k+1$ becomes the second bit plane of sprite $2k$.
PALETTE, ENTRY	Single sprite colour: entry 1-3 of the palette; entry 0 makes the sprite invisible (it still takes part in coincidence).
Pair colour	Pixel value = bit of sprite $2k + 2 \times$ bit of sprite $2k+1$, using entries 1-3 of the palette of sprite $2k$; the palette and entry fields of sprite $2k+1$ are ignored.

TABLE 2-4 - SPRITE COLOUR BYTE FIELDS

Both sprites of a pair keep their own position, name and pattern; the combined value exists where their pixels overlap. Their overlap does not set the coincidence flag C; overlaps with other sprites do. Priority is the priority of sprite $2k$.

Sprite selection is evaluated over line pairs: the vertical positions of sprites 0-15 are compared during even lines and those of sprites 16-31 during odd lines, and the selection is used for the next two lines. The eight-sprite limit applies to the sprites visible on either line of the pair; 5S and the sprite number in S0 report the ninth.

2.7 Scrolling and Scanline Interrupt

R9 and R10 scroll Graphics1X, Graphics2Fat, Bitmap and BitmapQ; R10 also scrolls the text modes. The memory access sequence does not change: the fetched tile column is $(c + 1 + R9/8)$ modulo 32 and the fine scroll selects the output tap of the pixel shift register. MASK blanks the first 8 pixels so that 32 fetched columns cover the visible picture for every fine scroll value. The world line selects name row, pattern row and third together.

The line counter is loaded from R12 at the first active line and whenever R12 is written. It is decremented at the end of every active line 0-191; when it would become negative, FL is set and the counter is reloaded. $R12 = 0$ interrupts on every line; $R12 = n$ interrupts every $n+1$ lines.

3. VDP INTERFACES AND OPERATION

3.1 VDP/VRAM Interface

The VRAM interface uses the TMS9918A terminals: AD0-AD7 (address and write data, AD0 = MSB), RD0-RD7 (read data), RAS, CAS and R/W, connected to eight 16K x 1 dynamic RAMs as described in MP010A Table 3-1.

3.1.1 VRAM Memory Types

The TMS9918B requires 16K x 1 dynamic RAMs with 120 ns row access time, 270 ns cycle time and page-mode operation, single +5 V types included. Examples: Fujitsu MB8118-12, Motorola MCM4517-12. 150 ns types do not meet the data setup requirement in any mode. R1 D0 (4/16K) must be 1. Appendix A gives the selection equations.

3.1.2 Address Multiplexing

A0-A6 (the most significant address bits) are output with RAS as row address and A7-A13 with CAS as column address. A page-mode cycle keeps the row and changes only the column, so its bytes lie in one 128-byte page; every table layout of Section 3.2 starts its multi-byte fetches at a multiple of the fetch length from a base that is a multiple of 128.

3.1.3 Memory Cycles

Every memory cycle reads or writes 1 to 4 bytes under one RAS pulse. Edges are placed on half-periods of the crystal (units of 46.56 ns). An n-byte cycle lasts $5n + 2$ units:

Signal	Timing (units from RAS falling edge)
RAS	low from 0 to $5n - 1$, precharge 3 units
CAS, byte k	low from $1 + 5k$ to $4 + 5k$; RD0-RD7 are latched on the rising edge
R/W (write, $n = 1$)	low from 2 to 4 (late write); AD0-AD7 carry the column address at the CAS falling edge, then the data, latched by the RAM on the R/W falling edge
Length	1 byte 7 units (326 ns), 2 bytes 12, 3 bytes 17, 4 bytes 22

TABLE 3-1 - MEMORY CYCLE STRUCTURE

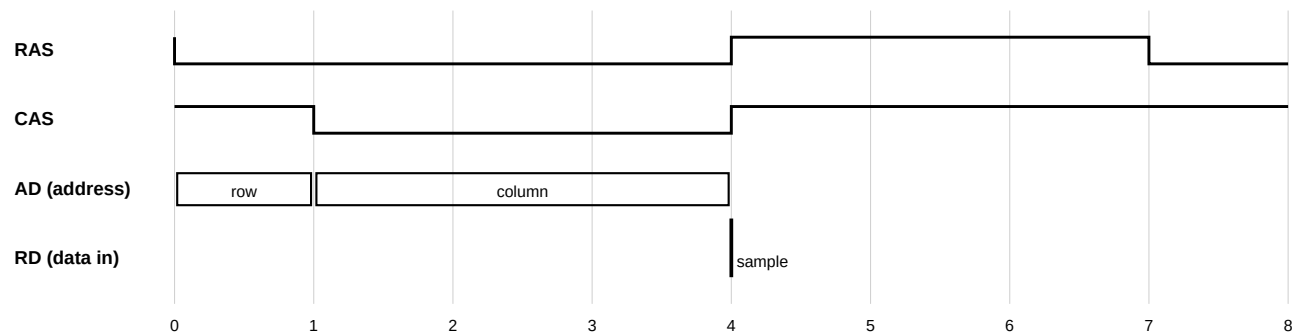


FIGURE 3-1 - SINGLE-BYTE READ CYCLE

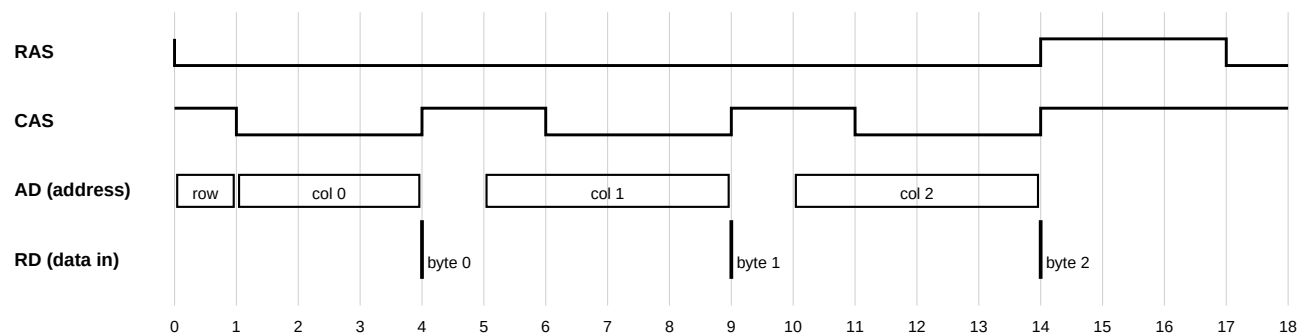


FIGURE 3-2 - THREE-BYTE PAGE-MODE READ CYCLE

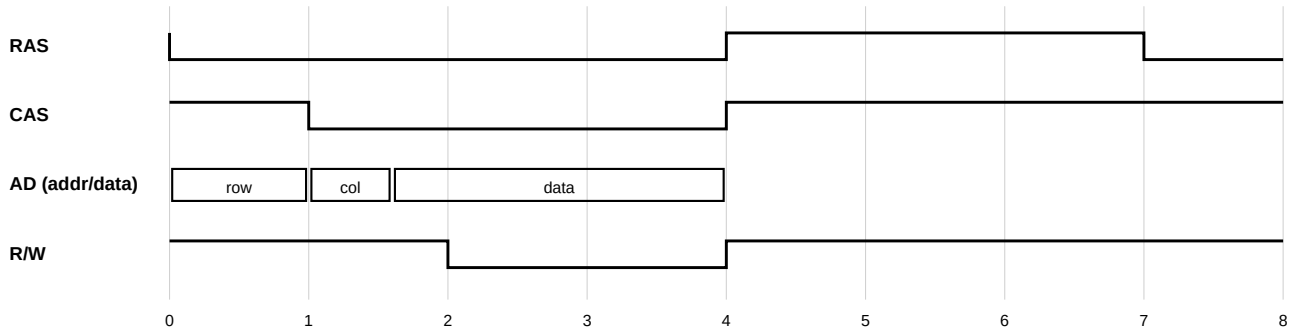


FIGURE 3-3 - WRITE CYCLE

3.1.4 Memory Access Sequence

In TMS9918A modes the VDP runs a single-byte cycle at the start of every TMS9918A slot followed by one idle unit, so the positions of the CPU access windows equal those of the TMS9918A. In extended modes each line follows one of the sequences of Appendix C. The positions refer to phase 0 of the line: unit = 2 x phase; TMS9918A slot *s* starts at unit $(8s + 1364) \bmod 1368$.

3.2 VRAM Memory Address Derivation

Table	Mode	Address of the first byte fetched	Bytes
Name entry	Graphics1X, Graphics2Fat	$(R2) \times 400h + 2 \times (\text{row} \times 32 + \text{column})$	2
Pattern row	Graphics1X	$(R4) \times 800h + 16 \times \text{name9} + 2 \times \text{row}$	2
Pattern row	Graphics2Fat	$(R4 \text{ D5}) \times 2000h + 16 \times (\text{third} \times 100h + \text{name AND mask}) + 2 \times \text{row}$	2
Bitmap	Bitmap	$(R4) \times 800h + 64 \times \text{line} + 4 \times (\text{column pair})$	4
Bitmap	BitmapQ	$(R4 \text{ D5}) \times 2000h + 1000h \times \text{block} + 64 \times (\text{line AND } 63) + 4 \times (\text{column pair})$	4
Palette map	Bitmap, BitmapQ	$(R2) \times 400h + 32 \times \text{row} + 2 \times (\text{column pair})$	2
Character entry	Text40X	$(R2) \times 400h + 4 \times (\text{row} \times 20 + \text{column pair})$	4
Name	Text64	$(R2) \times 400h + 64 \times \text{row} + 4 \times (\text{column group})$	4
Pattern	text modes	$(R4) \times 800h + 8 \times \text{name} + \text{row}$ (TMS layout; thirds with M3)	1
Sprite X, name, colour	extended graphics	$(R5) \times 80h + 4 \times \text{sprite} + 1$	3
Sprite Y	all	$(R5) \times 80h + 4 \times \text{sprite}$	1
Sprite pattern	all	$(R6) \times 800h + (\text{BANK}) \times 800h + 8 \times \text{name} + \text{row}$ (+16 right half)	1

TABLE 3-2 - EXTENDED MODE ADDRESS DERIVATION

Row, column, line and third refer to the world position after scrolling. In Graphics2Fat the mask is $(R4 \text{ D6 D7}) \times 100h + FFh$; in BitmapQ block *t* of the middle or bottom third is used when the corresponding R4 bit is 1, otherwise block 0. Addresses are modulo 4000h.

Mode	Tables (bytes)	Total
Graphics1X	patterns 8192, sprite patterns with BANK 4096, names 1536, SAT 128	13952
Graphics2Fat	patterns 12288, sprite patterns 2048, names 1536, SAT 128	16000
Bitmap, BitmapQ	bitmap 12288, sprite patterns 2048, palette map 768, SAT 128	15232
Text40X	patterns 2048, cursor patterns 256, entries 1920, SAT 128	4352
Text64	patterns 2048, cursor patterns 256, names 1536, SAT 128	3968

TABLE 3-3 - VRAM REQUIREMENTS OF THE EXTENDED MODES

3.3 Monitor Interfaces

The composite video output of the TMS9918B and the Y, R-Y and B-Y outputs of the TMS9928B/9929B have the levels and loads of MP010A Section 3.4. Luminance levels below full scale reduce luminance and colour difference in the same ratio above black level.

3.4 External VDP Operation

External VDP operation of the TMS9918B (EXTVDP, R0 D7) is that of the TMS9918A.

3.5 Oscillator and Clock Generation

The VDP uses a 10.738635 MHz crystal on XTAL1/XTAL2 or an external two-phase clock with the MP010A Section 5.4 limits (high and low pulse widths 42-52 ns, 42-52 ns phase delay from XTAL1 to XTAL2). Both clock phases time the memory cycle edges. CPUCLK (fext / 3) and GROMCLK (fext / 24) are unchanged.

3.6 VDP Terminal Assignments

TMS9918B				TMS9928B/TMS9929B			
RAS	1	40	XTAL1	RAS	1	40	XTAL1
CAS	2	39	XTAL2	CAS	2	39	XTAL2
AD7	3	38	CPUCLK	AD7	3	38	R-Y
AD6	4	37	GROMCLK	AD6	4	37	GROMCLK
AD5	5	36	COMVID	AD5	5	36	Y
AD4	6	35	EXTVDP	AD4	6	35	B-Y
AD3	7	34	RESET/SYNC	AD3	7	34	RESET/SYNC
AD2	8	33	VCC	AD2	8	33	VCC
AD1	9	32	RD0	AD1	9	32	RD0
AD0	10	31	RD1	AD0	10	31	RD1
R/W	11	30	RD2	R/W	11	30	RD2
VSS	12	29	RD3	VSS	12	29	RD3
MODE	13	28	RD4	MODE	13	28	RD4
CSW	14	27	RD5	CSW	14	27	RD5
CSR	15	26	RD6	CSR	15	26	RD6
INT	16	25	RD7	INT	16	25	RD7
CD7	17	24	CD0	CD7	17	24	CD0
CD6	18	23	CD1	CD6	18	23	CD1
CD5	19	22	CD2	CD5	19	22	CD2
CD4	20	21	CD3	CD4	20	21	CD3

FIGURE 3-4 - TERMINAL ASSIGNMENTS (IDENTICAL TO THE TMS9918A/9928A/9929A)

4. DEVICE APPLICATIONS

4.1 SC-3000 Class Systems

The TMS9918B replaces the TMS9918A or TMS9929A of an SC-3000 class system without board changes when the fitted VRAM meets Section 3.1.1. Boards produced in 1983 carry, among others, Fujitsu MB8118-12 (qualified) and Motorola MCM4517P15 (not qualified for the TMS9918B) dynamic RAMs. A board with 150 ns VRAM needs eight 120 ns page-mode RAMs in the same positions.

VDP terminal	Connected to
AD0	D input of RAM 0 (data only)
AD1 ... AD7	A6 ... A0 of all RAMs, and D input of RAM 1 ... RAM 7
RD0 ... RD7	Q output of RAM 0 ... RAM 7
RAS, CAS, R/W	RAS, CAS, WE of all RAMs

TABLE 4-1 - VDP TO VRAM CONNECTIONS (AS TMS9918A, MP010A TABLE 3-1)

4.2 Software Detection

Recommended detection sequence: (1) run F18A/PICO9918 detection first if supported; (2) write 5Ah twice to register 63 and restore R7; (3) write 01h to R8 (XE, bit D7; lands in R0 on other devices, restore R0); (4) write 01h to R15, read the status port twice and compare the second value AND 3Eh with 18h; (5) write 00h to R15 and restore R7.

4.3 Initialization

Extended modes are selected by writing XE and MX in R8, M1-M3 in R0/R1 and the table bases in R2, R4, R5 and R6, and by loading the palette. Register values, memory maps and example programs for every mode are given in the TMS9918B Programmer's Guide Supplement.

5. TMS9918B/9928B/9929B ELECTRICAL SPECIFICATIONS

Absolute maximum ratings, recommended operating conditions, electrical characteristics, CPU-VDP timing requirements and switching characteristics, video output characteristics and external clock requirements are those of the TMS9918A/9928A/9929A (MP010A Sections 5.1 to 5.5) unless listed below. Supply current is higher than the TMS9918A because of the added logic; its limits are to be determined.

5.1 Timing Requirements, VDP-VRAM Interface

Parameter	Description	MIN	NOM	MAX	Unit
tsu(D-CH)	RD0-RD7 setup time before CAS high	40			ns
th(CH-D)	RD0-RD7 hold time after CAS high	0			ns
ta(R)	VRAM access time from RAS (Appendix A)			120	ns
ta(C)	VRAM access time from CAS (Appendix A)			65	ns

TABLE 5-1 - VDP-VRAM TIMING REQUIREMENTS

5.2 Switching Characteristics, VDP-VRAM Interface

Parameter	Description	MIN	NOM	MAX	Unit
tc(1)	Memory cycle time, one byte (7 units)	321	326	331	ns
tc(P)	Page-mode byte period, CAS low to CAS low (5 units)	228	233	238	ns
tw(RL)	RAS low, one byte (4 units)	181	186	191	ns
tw(RL)n	RAS low, n bytes (5n - 1 units)		46.56 x (5n-1)		ns
tw(RH)	RAS precharge (3 units)	135	140	145	ns
td(RL-CL)	RAS low to CAS low (1 unit)	42	47	52	ns
tw(CL)	CAS low (3 units)	135	140	145	ns
tw(CH)P	CAS high inside a page-mode cycle (2 units)	88	93	98	ns
tw(CH)	CAS high between cycles (4 units)	181	186	191	ns
th(RL-RA)	Row address hold after RAS low	20			ns
tsu(CA-CL)	Column address setup before CAS low	0			ns
th(CH-CA)	Column address hold after CAS high	5			ns
tw(W)	R/W low, write cycle (4 units)	181	186	191	ns
tsu(D-WL)	Write data setup before R/W low	0			ns
th(RH-D)	Write data hold after RAS high	5			ns

TABLE 5-2 - VDP-VRAM SWITCHING CHARACTERISTICS (CL = 50 pF)

NOTE

Nominal values assume a 50 % clock duty cycle. MIN and MAX of odd unit counts follow the 42-52 ns clock phase limits; even unit counts equal whole clock periods.

5.3 CPU-VDP Interface

CPU-VDP timing requirements and switching characteristics are those of MP010A. Memory access delays are listed in Appendix B; palette and register writes are not delayed by memory cycles.

6. MECHANICAL DATA

The TMS9918B/9928B/9929B use the 40-pin plastic dual-in-line package of the TMS9918A/9928A/9929A (MP010A Section 6.1).

APPENDIX A - CHOOSING VRAM MEMORY

The VDP latches RD0-RD7 on the rising edge of CAS. The first byte of a cycle is latched 4 units (186.2 ns) after the falling edge of RAS; a following byte of a page-mode cycle is latched 3 units (139.7 ns) after the falling edge of its CAS. With $t_d(\text{RAS})$ and $t_d(\text{data})$ the board delays of the strobe and of the data:

$$186.2 \text{ ns} \geq \max[t_a(\text{R}), 46.6 \text{ ns} + t_a(\text{C})] + t_d(\text{RAS}) + t_d(\text{data}) + t_{su}(\text{D-CH})$$

$$139.7 \text{ ns} \geq t_a(\text{C}) + t_d(\text{CAS}) + t_d(\text{data}) + t_{su}(\text{D-CH})$$

With $t_{su}(\text{D-CH}) = 40 \text{ ns}$ the memories of Table A-1 allow the listed board delay. All other memory limits (cycle time, RAS and CAS pulse widths, precharge, page-mode timing, write timing) are met with the margins of Table A-2.

Part	$t_a(\text{R})$	$t_a(\text{C})$	Board delay allowed (first byte)	Board delay allowed (page byte)	Result
Fujitsu MB8118-12	120 ns	65 ns	26 ns	35 ns	qualified
Motorola MCM4517-12	120 ns	65 ns	26 ns	35 ns	qualified
Motorola MCM4517-15	150 ns	80 ns	-4 ns	20 ns	not qualified
TI TMS4116-20	200 ns	135 ns	-54 ns	-	not qualified

TABLE A-1 - VRAM ACCESS TIME BUDGET

Parameter (MB8118-12 limit)	VDP provides	Margin
tRC 270 ns	326 ns	56 ns
tRAS 140 ns	186 ns	46 ns
tRP 120 ns	140 ns	20 ns
tRCD 25-55 ns	47 ns	within
tCAS 65 ns	140 ns	75 ns
tCSH 120 ns	186 ns	66 ns
tRSH 85 ns	140 ns	55 ns
tCPN 55 ns	186 ns	131 ns
tCP 70 ns	93 ns	23 ns
tPC 145 ns	233 ns	88 ns
tWP 35 ns	93 ns	58 ns
tRWL 65 ns	93 ns	28 ns
tCWL 50 ns	93 ns	43 ns
tWCR 90 ns	186 ns	96 ns
tDH 35 ns	93 ns	58 ns

TABLE A-2 - MEMORY LIMITS AND MARGINS (NOMINAL CLOCK)

APPENDIX B - CPU TO VDP ACCESS TIMES

Condition	Mode	VDP delay	Time waiting for an access window	Total time	T-states @ 3.58 MHz
Active display area	Text (TMS)	2 us	0 - 1.1 us	2 - 3.2 us	12
Active display area	Graphics I, II (TMS)	2 us	0 - 6.0 us	2 - 8.0 us	29
Active display area	Multicolor (TMS)	2 us	0 - 1.5 us	2 - 3.5 us	13
Any time	Graphics1X, Graphics2Fat	2 us	0 - 3.0 us	2 - 5.0 us	18
Any time	Bitmap, BitmapQ	2 us	0 - 1.9 us	2 - 4.0 us	15
Any time	Text40X, Text40XQ	2 us	0 - 2.2 us	2 - 4.3 us	16
Any time	Text64, Text64Q	2 us	0 - 1.6 us	2 - 3.7 us	14
4300 us after vertical interrupt	All	2 us	0 us	2 us	8
Register 1 blank bit 0	All	2 us	0 us	2 us	8

TABLE B-1 - CPU TO VDP ACCESS TIMES

A continuous write loop never loses data when its period is at least the total time of the mode. In Bitmap, Text40X and Text64 modes an OUTI chain (16 T-states) is loss-free; in Graphics1X and Graphics2Fat an OTIR (21 T-states per byte) is loss-free.

APPENDIX C - MEMORY ACCESS SEQUENCES

Offsets are in units (46.56 ns) from the first unit of a group; figures in parentheses are bytes fetched. The complete position of every cycle of every line is listed in the TMS9918B model report (calendars.md).

C.1 Graphics1X and Graphics2Fat - two cells per 64 units from unit 212

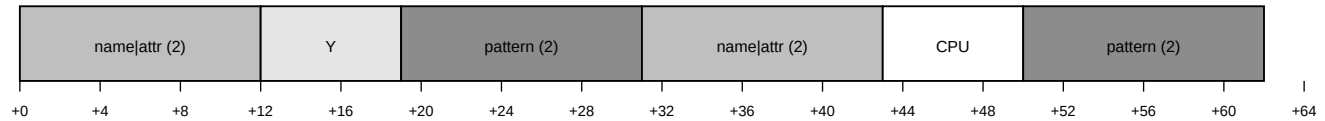


FIGURE C-1 - TILE MODE CELL PAIR

C.2 Bitmap and BitmapQ - two cells per 64 units from unit 212



FIGURE C-2 - BITMAP MODE CELL PAIR

C.3 Text40X - two characters per 48 units from unit 236

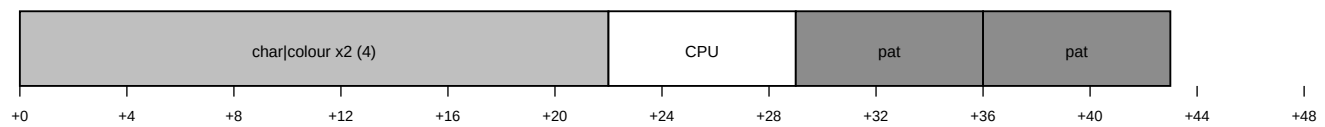


FIGURE C-3 - TEXT40X CHARACTER PAIR

C.4 Text64 - four characters per 64 units from unit 212

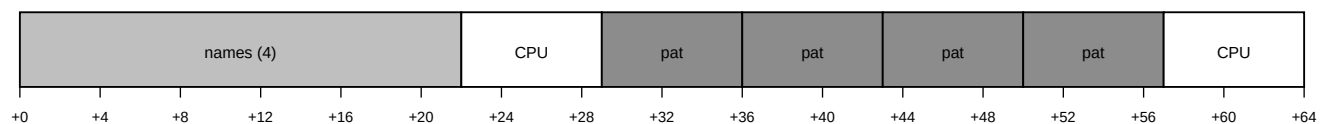


FIGURE C-4 - TEXT64 CHARACTER GROUP

C.5 Horizontal blanking of the extended graphics modes (units 1236-1579)

One CPU cycle; four groups of two sprites a, b of 76 units: sprite X, name and colour of a (3), of b (3), CPU, left half a, right half a, left half b, right half b, CPU; four CPU cycles. Text modes fetch the cursor sprites 0 and 1 in the same way and use the rest of the blanking for CPU cycles.

Sequence	CPU cycles per line	Sprites per line	Worst CPU delay
Graphics1X, Graphics2Fat	29	8	18 T
Bitmap, BitmapQ	45	8	15 T
Text40X	67	cursor	16 T
Text64	70	cursor	14 T

TABLE C-1 - SEQUENCE SUMMARY